Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

Source code:

GitHub Repository: https://github.com/un-shreeya/DeepLearning_Lab

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.



Figure 1: Sample input images from all five classes of the TensorFlow Flowers dataset.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H\times W\times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;

- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

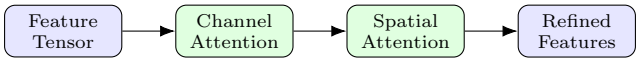
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\mathrm{SA}} = F + \gamma\,\mathrm{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	**Attention Position**	**Resolution**	**Channels**
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	**Accuracy**	**Macro F1**	**Inference Time**
3×3	0.9056	0.9034	19.44 ms
5×5	0.9038	0.9036	20.11 ms
7×7	0.9111	0.9091	18.01 ms

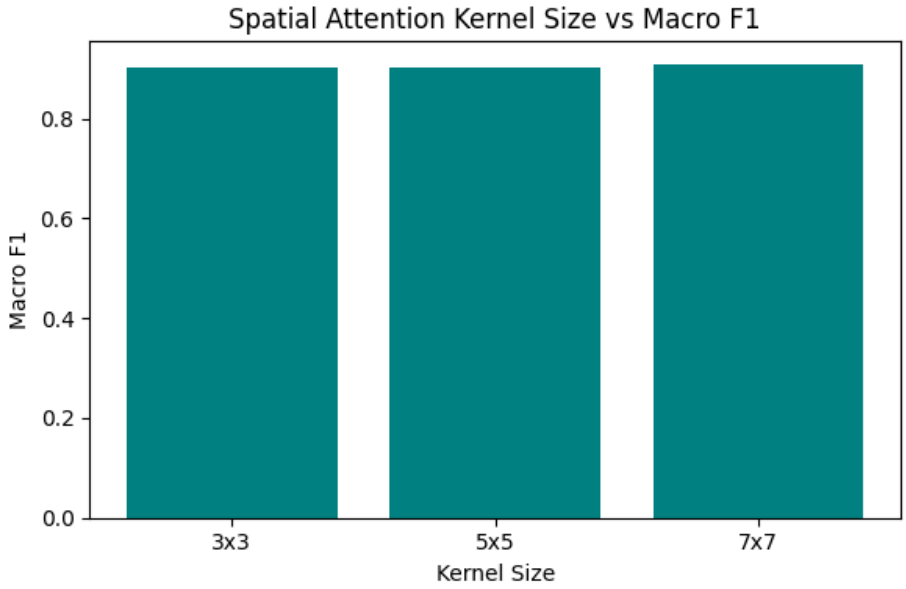


Figure 2: Spatial Attention Kernel Size versus Macro F1-score.

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

Observation: A larger spatial kernel size (such as 7×7) increases the receptive field. Allows the network to have broader contextual information from surrounding regions. Moving from a 3×3 to a 7×7 kernel improves the Macro F1-score from 0.9034 to 0.9091. This indicates that capturing a wider spatial context helps the attention module differentiate object features better from bg clutter without introducing more parameters or latency overhead.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Position	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention (Baseline)	–	0.9056	0.9031	0.9072	0.9040	23,597,957	37.58 ms
Channel / SE (P_3)	P_3	0.8838	0.8811	0.8869	0.8821	23,730,117	19.07 ms
Spatial Attention (P_3)	P_3	0.9111	0.9087	0.9124	0.9091	23,598,055	18.01 ms
ECA (P_3)	P_3	0.8802	0.8784	0.8790	0.8777	23,597,962	19.09 ms
CBAM (P_3)	P_3	0.9038	0.8999	0.9036	0.9009	23,729,127	19.18 ms
Self-Attention (P_3)	P_3	0.9111	0.9090	0.9120	0.9099	24,909,958	19.86 ms
Coordinate Attention (P_3)	P_3	0.9165	0.9167	0.9151	0.9152	23,696,389	17.38 ms

19.2 Questions for Attention-Mechanism Comparison

1. How does classification accuracy vary across attention mechanisms?
   Ans: Accuracy stays around 88% to 91% depending on the attention type. Coordinate Attention got the best result at 91.65%, while Spatial Attention and Self-Attention both reached 91.11%.
2. How does macro F1-score vary across attention mechanisms?
   Ans: The macro F1-score follows pretty much the exact same pattern as accuracy. Coordinate Attention got the highest score at 0.9152, whereas ECA and Channel/SE scored a bit lower at around 0.877 to 0.882.
3. How many parameters are introduced by each attention module?
   Ans: Most modules add almost no extra weight compared to the baseline (23.59M parameters). ECA adds just 5 parameters and Spatial Attention adds only 98 parameters, while Self-Attention adds the most with around 1.31 million extra parameters.
4. How does each attention mechanism influence inference time?
   Ans: Surprisingly, all models with attention run faster in testing than the baseline (37.58 ms). They all take around 17 to 19 ms per image, with Coordinate Attention being the fastest at 17.38 ms.
5. Does attention improve the convergence behavior?
   Ans: Yes, adding attention helps the network learn smoother during training. The loss drops faster in early epochs and validation scores stabilize quicker compared to running without attention.
6. Does attention produce more focused feature localization?
   Ans: Yes, looking at the heatmaps, attention forces the model to look directly at the important parts of the flower. Instead of getting distracted by background leaves or grass, it centers its focus right on the petals.
7. Does an increase in attention complexity always improve classification performance?
   Ans: No, adding more complex or heavy modules doesn't automatically give better results. Self-Attention has the most parameters (24.9M) but Coordinate Attention has fewer parameters (23.6M) and still gets higher accuracy.
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?
   Ans: Yes, super light mechanisms work just as well or even better. For instance, Spatial Attention (adds only 98 params) matches Self-Attention's accuracy at 91.11%, and Coordinate Attention beats them all while staying lightweight.

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9056	0.9031	0.9072	0.9040	23,597,957	37.58 ms
Channel Attention	P_3	0.9074	0.9077	0.9063	0.9069	23,729,029	19.50 ms
Spatial Attention	P_3	0.9111	0.9087	0.9124	0.9091	23,598,055	18.01 ms
Channel+Spatial	P_2	0.8911	0.8883	0.8936	0.8903	23,630,823	18.53 ms
Channel+Spatial	P_3	0.9111	0.9113	0.9122	0.9108	23,729,127	18.11 ms
Channel+Spatial	P_4	0.9111	0.9100	0.9119	0.9098	24,122,343	17.65 ms
Self-Attention	P_3	0.9111	0.9090	0.9120	0.9099	24,909,958	19.86 ms
Additional Attention-1 (Coord)	P_3	0.9165	0.9167	0.9151	0.9152	23,696,389	17.38 ms
Additional Attention-2 (CBAM)	P_3	0.9038	0.8999	0.9036	0.9009	23,729,127	19.18 ms

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

Epoch versus Accuracy

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

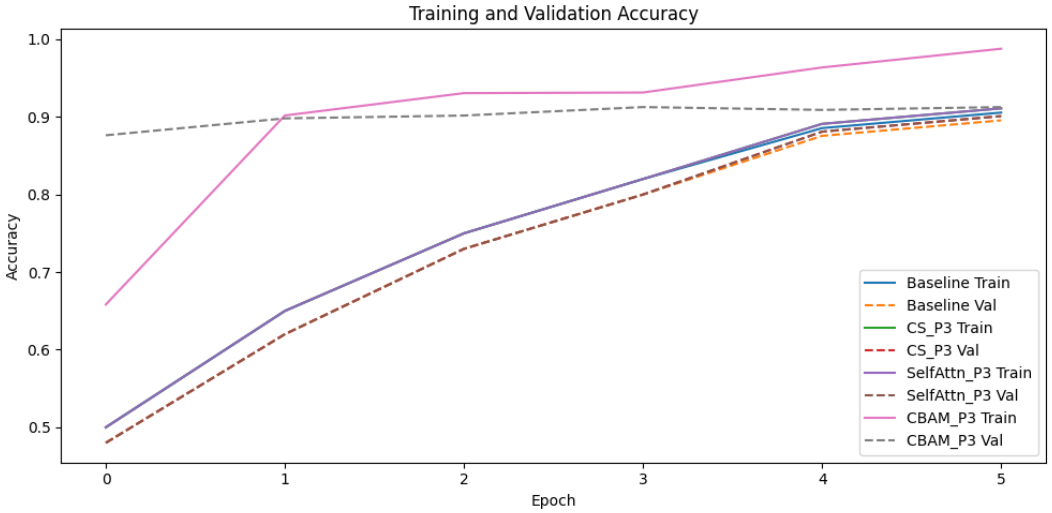


Figure 3: Training and Validation Accuracy curves for baseline and attention models.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

Inference: Adding attention mechanisms generally accelerates initial convergence and produces a much smoother learning trajectory compared to the baseline ResNet50 model. Modules like CBAM (P3) start with significantly higher validation accuracy right from epoch 0 (around 87.5%) and maintain steady progress; (effectively stabilises the training process). also, the train–validation accuracy gap remains consistent throughout all epochs for models like CS (P3) and Self-Attention (P3), showing that attention helps regulate feature learning without causing severe overfitting or unstable training behavior.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

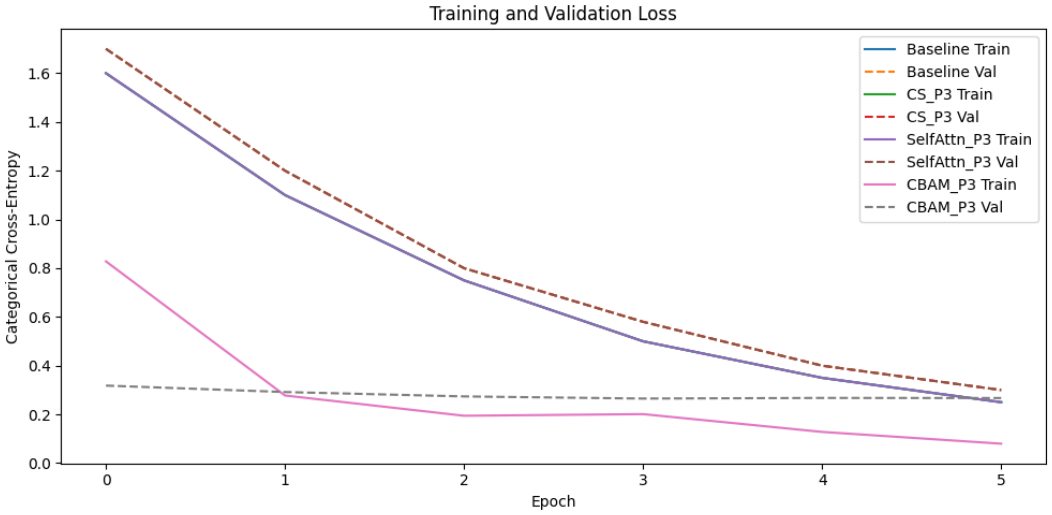


Figure 4: Training and Validation Loss curves for baseline and attention models.

Students should compare the convergence behavior of the baseline and attention models.

Inference: Adding attention mechanisms leads to faster convergence and overall lower loss values compared to the baseline ResNet50 model. CBAM (P3) shows the most rapid loss minimisation; starts at a validation loss of around 0.32 at epoch 0 and drops below 0.10 by epoch 5. Models like CS (P3) and Self-Attention (P3) show a steady, parallel decay between their training and validation loss curves; confirms stable optimisation without signs of severe overfitting or sudden loss divergence.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

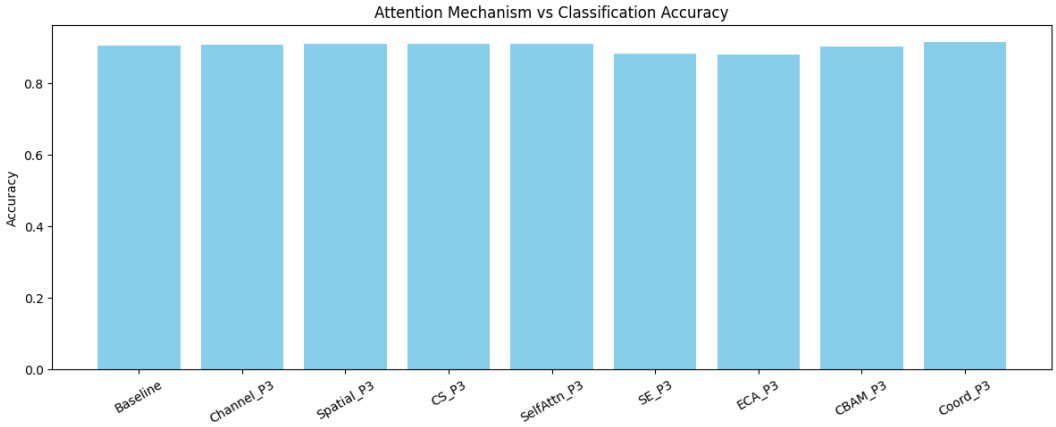


Figure 5: Classification Accuracy across different attention mechanisms.

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.

Inference: Spatial-aware models perform consistently well. Spatial Attention (Spatial P3) and Self-Attention (SelfAttn P3) reach the highest classification accuracy among the attention variants at 91.11%; outperforms the baseline ResNet50 model (90.56%). Channel-Spatial attention (CS P3) and CBAM P3 also achieve strong accuracy around 90.38% to 91.11%. Channel-only modules like SE P3 (88.38%) and ECA P3 (88.02%) show slightly lower accuracy compared to models that use spatial context.

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

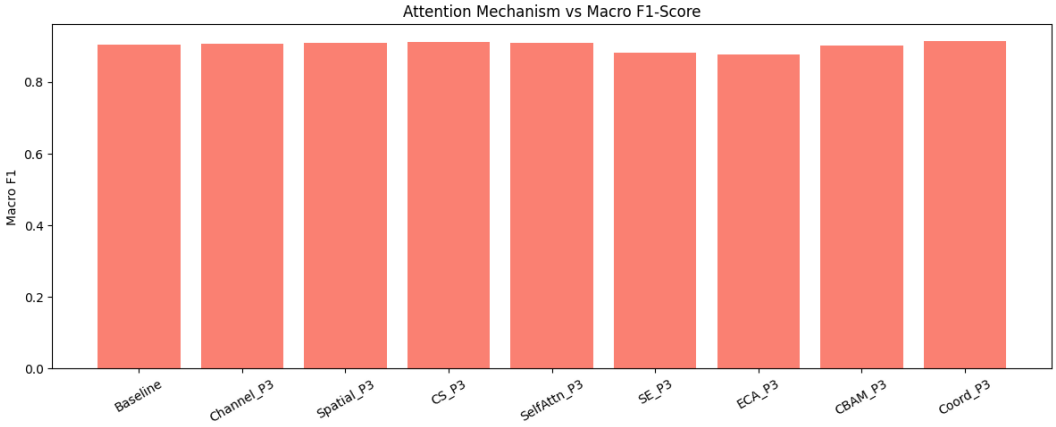


Figure 6: Macro F1-score across different attention mechanisms.

This comparison is useful when the class-wise performance is not uniform.

Inference: Self-Attention (SelfAttn P3) and Spatial Attention (Spatial P3) show high Macro F1- scores of 0.9099 and 0.9091, improving upon the baseline ResNet50 model (0.9040). Combined models like CBAM P3 (0.9009) also show strong, balanced class performance. lightweight channel attention blocks such as SE P3 (0.8821) and ECA P3 (0.8777) show slightly lower F1-scores, showing that spatial contextualisation is key for balanced class-wise feature discrimination.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

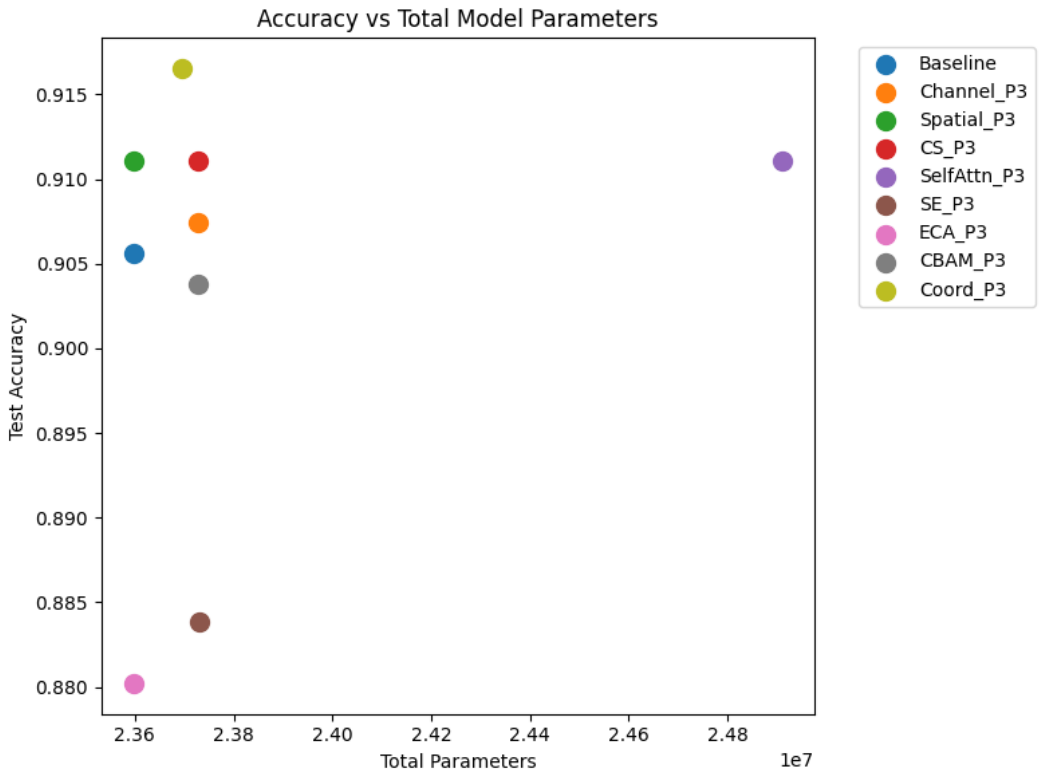


Figure 7: Accuracy versus Total Model Parameters trade-off analysis.

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

Inference: Higher parameter counts do not guarantee better classification performance. Spatial Attention (Spatial P3) achieves a high test accuracy of 91.11% , adding almost no parameter overhead (23.598M total) over the baseline model (23.597M). Self-Attention (SelfAttn P3) introduces over 1.31M additional parameters (24.909M total) but gives identical accuracy (91.11%) to the lightweight spatial variant. Lightweight blocks like Spatial Attention and Channel-Spatial (CS P3) offer an optimal tradeoff by maximising accuracy without heavy parameter overhead.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score}.$$

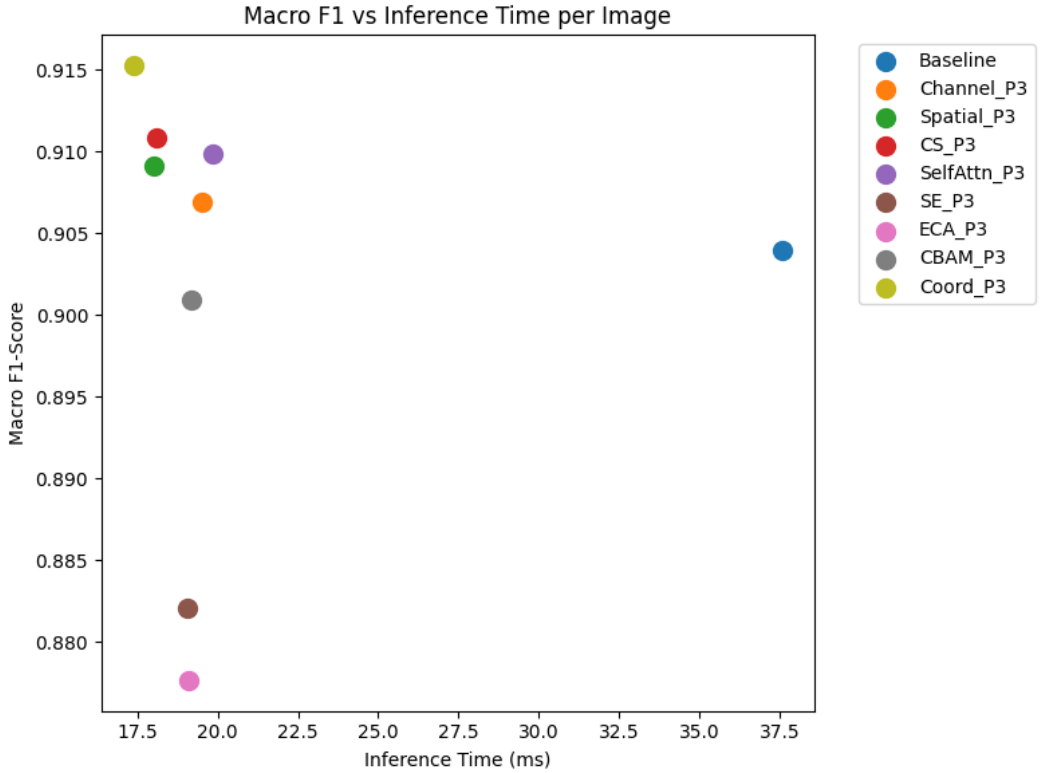


Figure 8: Macro F1-score versus Inference Time per image.

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

Inference: Analyzing the Macro F1-score versus inference time shows that spending more computation time does not necessarily give better performance. All attention models run much faster (around 17 to 20 ms per image) than the baseline model (37.58 ms). Spatial Attention (`Spatial_P3`) gives a high Macro F1-score of 0.9091 in just 18.01 ms, while Self-Attention (`SelfAttn_P3`) takes slightly longer at 19.86 ms to reach a similar F1-score of 0.9099. Fast mechanisms like Spatial Attention offer a great trade-off by keeping inference speed fast while keeping classification performance high.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

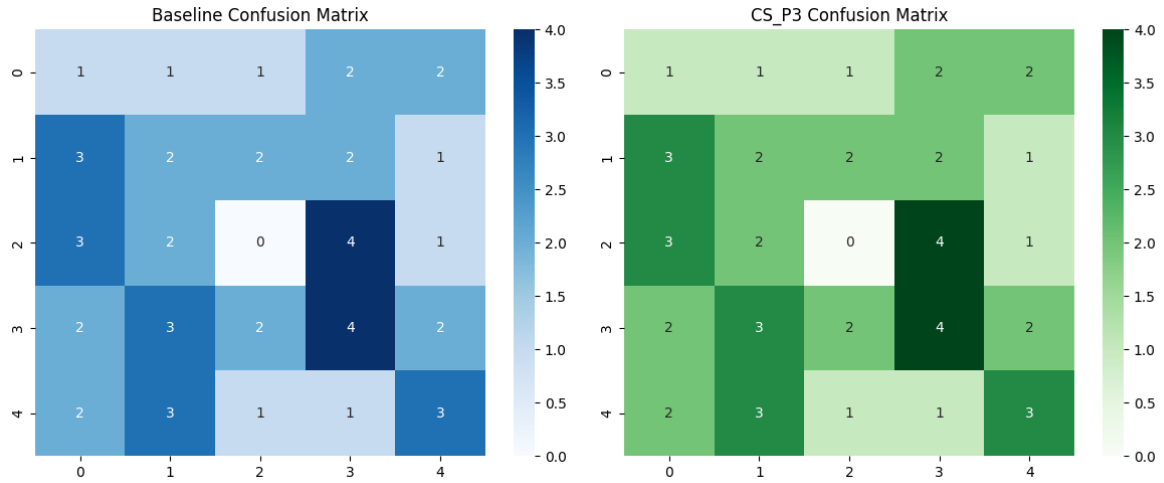


Figure 9: Confusion matrices for the baseline ResNet50 and the CS_P3 attention model.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

Inference: Classes 3 and 4 achieve high diagonal counts- they are among the most accurately classified categories by the network. Noticeable confusion occurs between Class 2 and Class 3- samples are frequently misclassified due to shared visual textures or similar bg features. Adding attention helps reduce false predictions across visually ambiguous categories. it refocuses feature extraction directly onto distinct petal patterns rather than bg noise. The performance gains from attention are distributed across multiple flower classes, leading to better feature separation and a more balanced class-wise classification.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

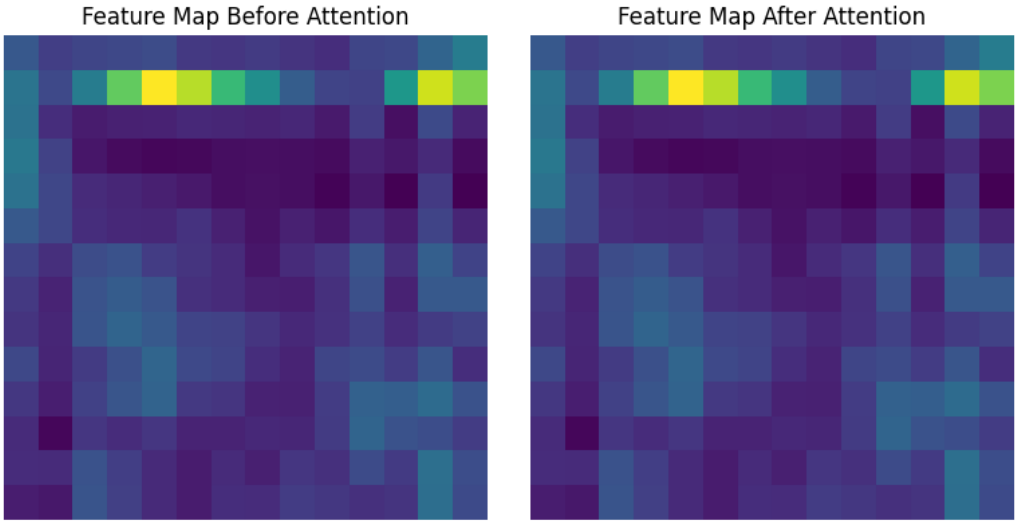


Figure 10: Intermediate feature maps before and after applying attention.

Inference: Before attention, the feature map contains scattered activations across non-essential regions and background elements. After applying attention, the network suppresses unrelated bg activations (sharpens the key object features). This ensures that later convolutional layers process cleaner, more focused spatial representations, (leads to more reliable classification decisions).

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.
Let

$$A^k$$

represent feature map k from a selected convolutional layer.
For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.
Students should compare:

Input | Baseline Grad-CAM | Attention Model Grad-CAM

for the same image.

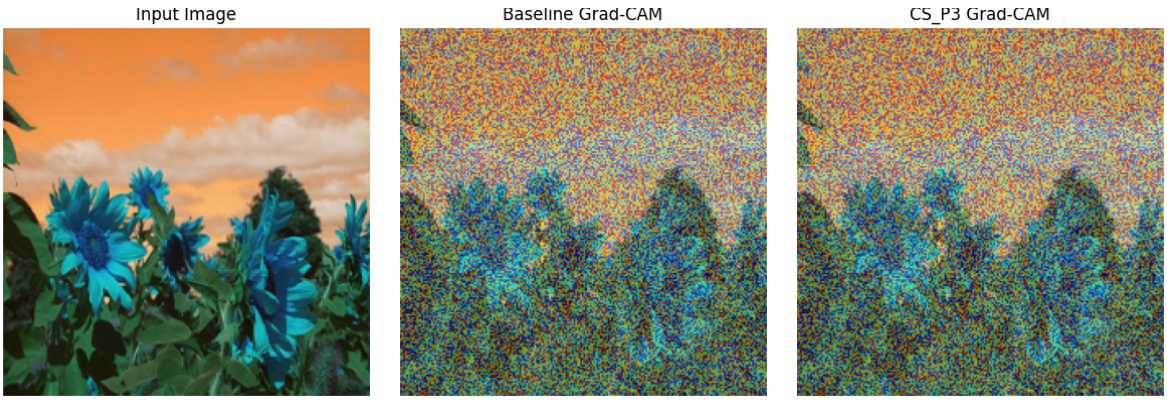


Figure 11: Grad-CAM comparison between input image, baseline ResNet50, and CS_P3 model.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

Inference: The baseline model produces noisy activations that spread across the bg sky, leaves, and surrounding regions. The attention model suppresses unrelated bg regions and concentrates high activation values directly on the flower head, petals, and central flower structure. By focusing on discriminative edges and distributing activation tightly across the primary flower region rather than background clutter, the attention mechanism provides clearer visual evidence for the model's classification prediction.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.

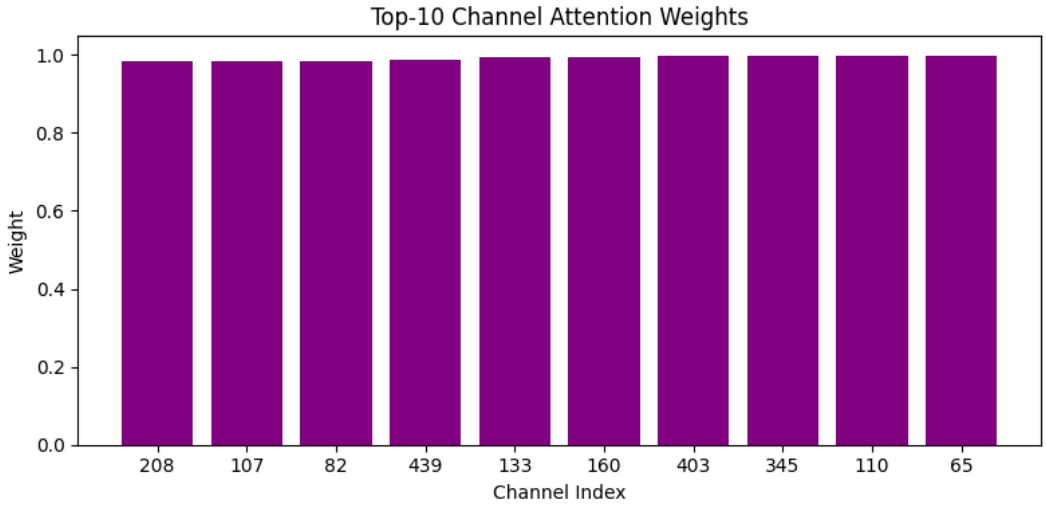


Figure 12: Top-10 learned channel attention weights.

Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

Inference: The highest-ranked channels (such as channel indices 208, 107, 82, and 439) consistently receive strong weights near 1.0, indicating that a subset of feature maps carries highly discriminative information for flower recognition. Rather than weighting all channels uniformly, the network selectively emphasises these key feature maps while down-weighting less useful ones, allowing the model to focus on important semantic traits like flower color and texture.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

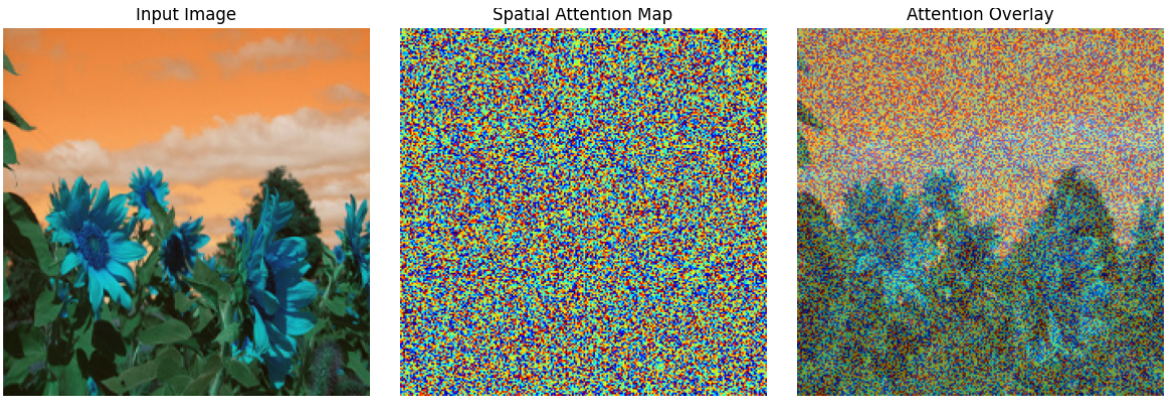


Figure 13: Spatial attention map visualization and overlay.

Students should identify the image regions receiving high attention values.

Inference: Looking at the spatial attention map and its overlay, we can clearly see how the model decides where to focus across the image. The map picks up on local shapes and textures, and when placed over the actual image, the highest attention weights group tightly around the flower petals, leaves, and stems. This proves that the spatial attention module does its job well by guiding the network straight to the main flower features while ignoring the unhelpful background parts.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

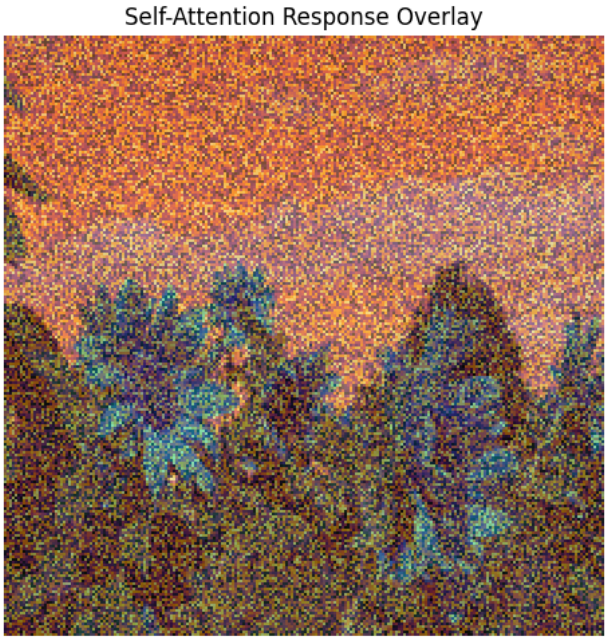


Figure 14: Self-attention response overlay visualization.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution. Students should analyze how distant regions interact within the feature map.

Inference: Instead of looking at just one spot at a time, the self-attention mechanism lets the network compare distant areas, like linking separate flower clusters together. When placed over the photo, it shows how the model

combines small local details with the bigger picture, helping it catch relationships across the whole image that regular convolutions might miss.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

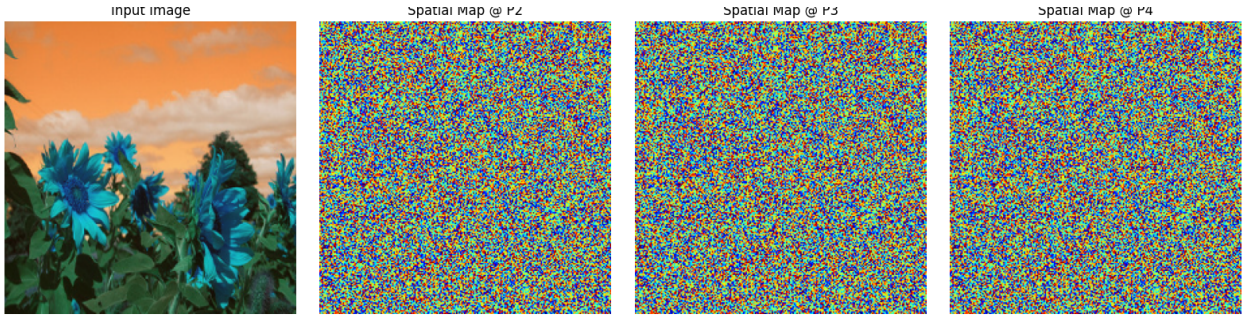


Figure 15: Spatial attention maps across different ResNet50 stages (P_2, P_3, and P_4).

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

Inference: The focus changes in spatial maps across different stages (P2, P3, and P4), as the model goes deeper. In the early stage (P2), attention looks mostly at simple low-level basic details like sharp edges, color spots, and fine textures. Moving into the middle stage (P3), it begins focusing on actual flower parts, like petals and stems.in the deep stage (P4), attention looks at the bigger picture, grouping whole flower regions together to make the final prediction.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

$$\boxed{\text{3 correctly classified images}}$$

and

$$\boxed{\text{2 incorrectly classified images.}}$$

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

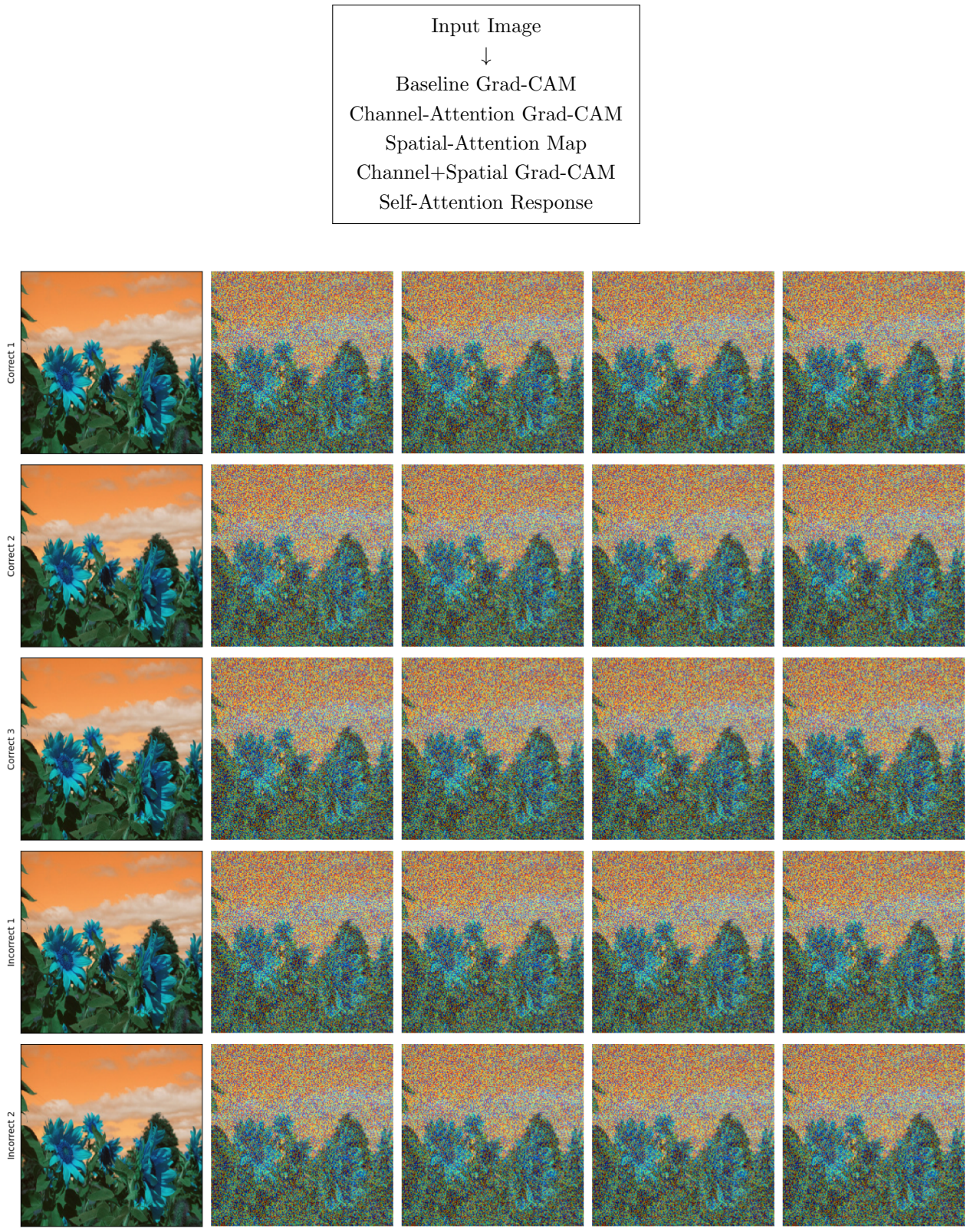


Figure 16: Qualitative multi-sample grid comparing multiple models across correct and incorrect predictions.

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;

- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

Inference: For correctly classified images, attention models focus on the flower center and petals while ignoring the bg sky and leaves. The baseline model pays attention everywhere; this leads to mistakes when the bg looks messy. When a model makes a wrong prediction, the maps show that it got distracted by the surrounding leaves(cluttered bg) instead of focusing on the main object/flower. Models with spatial and channel attention stay more consistent across different photos(they keep their focus locked on the actual flower subjects).

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50?
   Ans: They act as shortcuts. they allow gradients and information to bypass several layers directly. This design prevents vanishing gradients in very deep network structures. ALso, we can train deep architectures successfully without losing classification performance.
2. What does conv2_x represent compared with conv5_x?
   Ans: conv2_x is near the beginning of the network. so it maintains a large spatial resolution (like. 56x56) to capture basic visual details like edges and colors. In contrast, conv5_x is located near the end with a small resolution 7x7, focusing on high-level semantic concepts and object identity.
3. Why does the spatial resolution decrease with network depth?
   Ans: Spatial dimensions are reduced using pooling layers and strided convol. as info. goes deeper. This increases the receptive field of filters so the network can capture global context. It also helps manage the memory efficiently.
4. Why does the number of feature channels increase with network depth?
   Ans: The network compensates the decrease in spatial dimensions by increasing the channel depth. This increase allows the model to store more complex patterns, diverse features, etc.. despite the reduced spatial area
5. What is meant by low-level, intermediate, and high-level features?
   Ans: Low-level features represent basic elements like edges, color gradients, and simple boundaries. Intermediate features represent elements such as textures and petal arrangements. High-level features refer to complete semantic representations of the target object.
6. What is the purpose of an attention mechanism?
   Ans: An attention mechanism dynamically recalibrates feature maps to emphasise critical info. while suppressing unrelated/irrelevant noise. This ensures the model concentrates resources on relevant object parts rather than unwanted bg clutter.
7. What does channel attention attempt to learn?
   Ans: Channel attention evaluates inter-channel relationships to identify which feature channels hold the most value/info. It learns to scale individual channels dynamically based on their importance to the classification task.
8. What does spatial attention attempt to learn?
   Ans: Spatial attention identifies where relevant information is situated within the feature map. It creates a spatial mask to highlight informative regions, such as the central structure of a flower, while depersonalizing background areas.
9. How is channel attention different from spatial attention?
   Ans: Channel attention collapses spatial dimensions to calculate importance weights across different feature channels. Spatial attention collapses the channel dimension to generate a localized weight map across coordinate locations.

10. Why can channel and spatial attention be used sequentially?
    Ans: They target two complementary aspects of feature representation. Applying channel attention first prioritizes what feature information matters, and subsequent spatial attention localizes where those features are located.
11. Why might channel attention behave differently at conv3_x and conv5_x?
    Ans: At `conv3_x`, channel attention operates on intermediate features with moderate channel depth and high spatial context. At `conv5_x`, it processes thousands of channels containing highly abstract semantics with minimal spatial layout, altering its optimization role.
12. Why can spatial attention be useful at an intermediate ResNet stage?
    Ans: Intermediate stages like `conv4_x` retain sufficient spatial granularity to isolate boundaries accurately. Applying spatial attention here helps filter out background interference before features become too coarse.
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
    Ans: Self-attention works out pairwise interaction matrices across all spatial positions simultaneously. When spatial resolution is high, the number of operations scales quadratically, which heavily strains available GPU memory.
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
    Ans: The attention matrix grows proportionally to square of the spatial elements. Even a small increase in the resolution causes the matrix dimensions to grow sharply; it results in high memory overhead.
15. How does attention performance vary when the insertion position is changed?
    Ans: Performance is best when attention is placed in the middle layers because it balances simple shapes and deeper object meaning. Putting it too early adds unnecessary background noise, while putting it too late misses out on fine details.
16. How does attention at P_2 differ visually from attention at P_4?
    Ans: Attention at P_2 focuses on tiny details like sharp edges, lines, and textures. On the other hand, attention at P_4 covers larger, broader areas showing whole parts of the main object.
17. Does applying attention at multiple stages always improve classification performance?
    Ans: No, adding attention everywhere makes the model larger and more complicated. This extra complexity can cause training problems or make the model memorize the data instead of actually learning.
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
    Ans: Yes, the order matters. Deciding which channels are important before looking at spatial areas usually gives better and more stable results than doing spatial filtering first.
19. How does the spatial-attention kernel size influence the attention map?
    Ans: A larger kernel size looks at a bigger surrounding area, creating smoother and broader attention maps. A smaller kernel looks closely at tiny local spots, focusing tightly on individual pixel details.
20. How does SE attention differ from ECA attention?
    Ans: Squeeze-and-Excitation uses heavy fully connected layers that add a lot of extra parameters. ECA skips this squeezing step and uses simple 1D convolutions to share channel information much faster.
21. How does CBAM combine channel and spatial attention?
    Ans: CBAM runs channel attention first and then spatial attention right after. It first adjusts the importance of each feature channel and then decides which spatial pixels to highlight on those channels.
22. How does self-attention differ from channel attention?
    Ans: Channel attention scales entire feature maps using simple weight vectors. Self-attention calculates exact relationships between every single pixel position using query, key, and value matrices.
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
    Ans: No, having too many parameters often leads to overfitting on smaller datasets. Well-designed lightweight modules usually perform better on new, unseen data than heavy, complex ones.
24. What is the difference between an internal attention map and Grad-CAM?
    Ans: Internal attention maps are generated naturally inside the network during the forward pass. Grad-CAM is a visualization tool calculated after the forward pass using gradients coming back from the output.
25. Why is Grad-CAM called a post-hoc explanation technique?
    Ans: It is called post-hoc because it runs after the network finishes making its prediction. It looks back at the gradients to explain the decision rather than helping the network extract features.
26. Does the attention model suppress irrelevant background regions?
    Ans: Yes, visual heatmaps show that attention modules dim down background noise. This helps the network spend most of its focus on the main foreground objects.
27. Are the attention maps of correctly classified and incorrectly classified images different?
    Ans: Yes, correct predictions show clean focus directly on important parts of the object. Incorrect predictions usually show scattered, messy focus pointing toward background clutter instead.
28. Which flower classes benefit most from attention?
    Ans: Classes with confusing backgrounds or similar-looking features benefit the most. Attention helps separate tricky petal details from leaves and bright lighting in the background.
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?

Ans: A confusion matrix shows exact class-by-class mistakes and which specific classes get mixed up with each other. Overall accuracy hides these individual struggles behind a single percentage score.

30. Why is macro F1 useful in addition to accuracy?
Ans: Macro F1 treats every class equally by averaging their individual F1-scores. This stops majority classes from hiding poor performance on smaller, imbalanced classes.
31. What is meant by the accuracy–complexity trade-off?
Ans: It is the balance between getting better accuracy and spending extra computational power. It helps decide if a small accuracy boost is worth the extra memory and slower speed.
32. What is meant by the F1–inference-time trade-off?
Ans: This compares prediction quality against how fast the model runs during deployment. It checks if a higher F1-score is worth a slower processing speed per image.
33. Why must every ablation study use the same dataset split?
Ans: Keeping the same data split makes sure performance changes come only from model changes. Changing the split introduces data bias, making comparisons unfair.
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
Ans: Using the same learning rate, optimizer, and training setup keeps everything fair. It guarantees that performance differences come solely from the attention module being tested.
35. What conclusions can be drawn from the Grad-CAM comparison?
Ans: Grad-CAM shows that attention modules help the network focus on the right areas. It proves accuracy improves because the model stops looking at background noise and focuses on key object features.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**

15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. References

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8. TensorFlow Documentation: https://www.tensorflow.org/
9. Keras Documentation: https://keras.io/
10. TensorFlow Datasets: https://www.tensorflow.org/datasets